Kru — Project Plan & Milestones

Nov 1, 2026 · @Amin Haiqal

Overview

Kru is a headless, local-first engine that runs AI agent CLIs (Claude Code, Codex, Gemini, Cursor and others) as persistent teammates. Every agent gets a durable mailbox, a workspace and a thread that survives provider switches and crashes. Work is delivered one turn at a time, and agents message each other through the same mailbox.

Pitch: *Give every agent a mailbox. Kru delivers the work and never loses a message.*

The name: *kru* is Malay for "crew" — a crew of agents that hand work to each other. One syllable, easy to say, easy to type as a command.

Who uses it

- Developers who want several agent CLIs to work on one codebase, each with its own workspace and memory.
- Teams that put agents behind their own tools: a Slack bot, a CI job, an internal dashboard.
- Builders who need an engine they can embed, not a desktop app.

How it differs from OpenBot

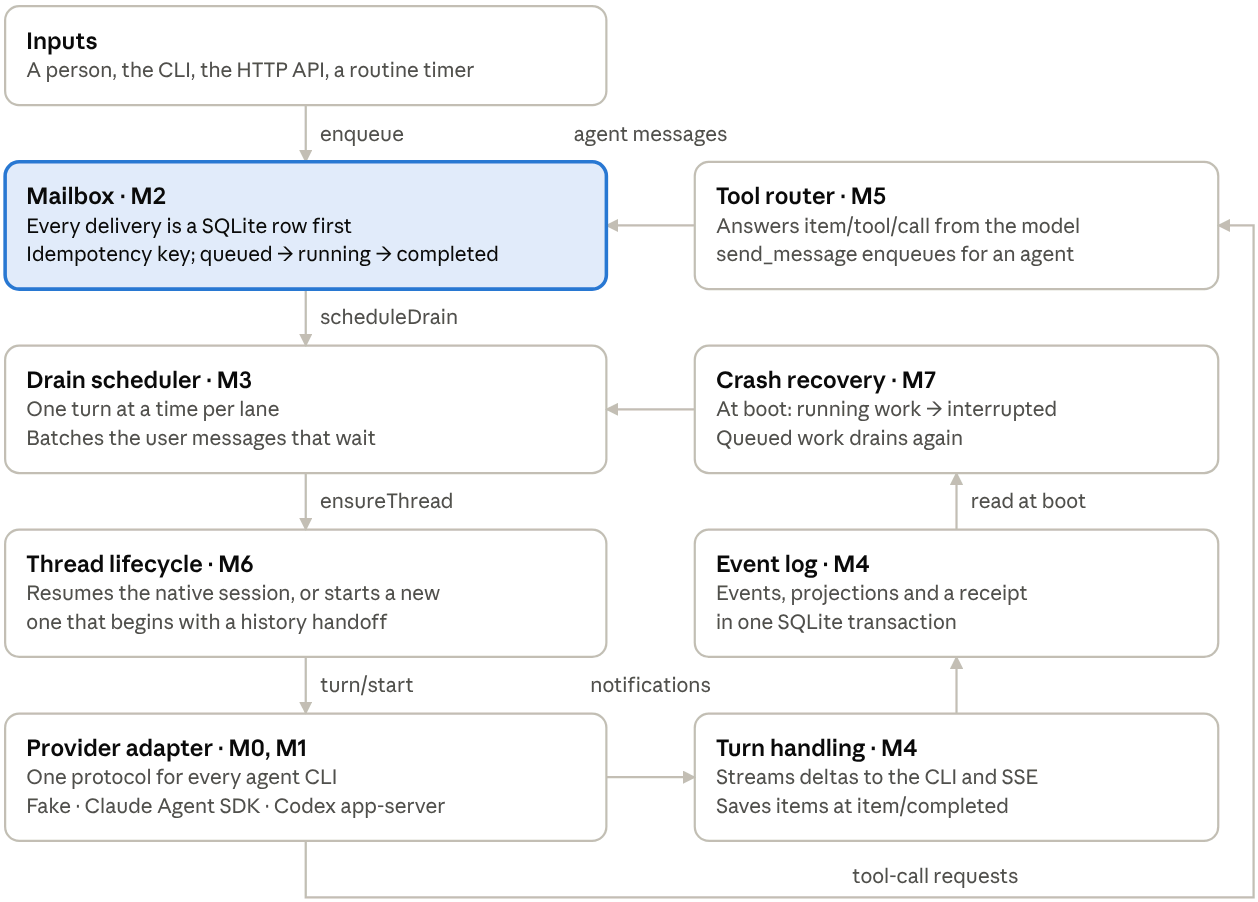
	OpenBot	Kru
Form	Electron desktop app	Daemon, CLI, HTTP + SSE API, TypeScript library
Audience	End users	Developers and integrators
Scope	Browser, mobile, voice, team servers	The core engine only
License	PolyForm Noncommercial 1.0.0	MIT (allows commercial use)
Code	—	Written from scratch from the ideas; no copied code

The license row is why the last row matters: OpenBot's code cannot go into an MIT project, so study a file, close it, then write your own version.

Architecture

Work flows down the left column and results flow back up the right. Agent messages loop back into the same mailbox, so there is only one execution path.

Every piece of work enters through the mailbox



Kru message lifecycle · 9 parts, labelled by milestone

Core concepts

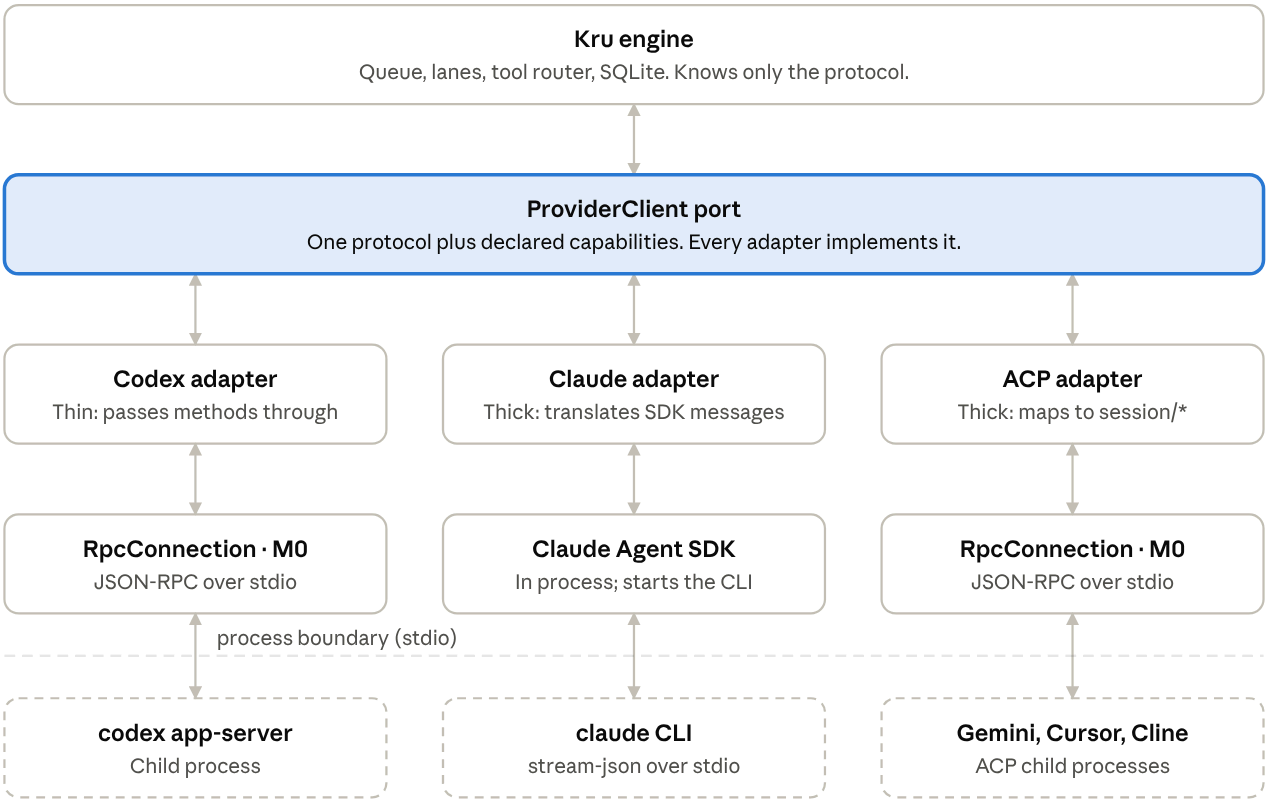
- **Delivery:** one unit of work for one agent, stored before anything runs.
- **Lane:** a queue that runs one turn at a time — an agent's main chat, or one of its side threads.
- **Thread vs provider session:** the thread is Kru's and permanent; the session is the CLI's and disposable.
- **Handoff:** a size-limited summary of recent history, prepended to the first turn of a new session.

- **Turn:** one `turn/start` to `turn/completed` exchange with a provider.

Provider layers: ports and adapters

Kru's engine depends on one interface, never on a specific agent. This is hexagonal architecture, also called ports and adapters: the `ProviderClient` interface is the port, and each agent CLI plugs in through its own adapter. Adding an agent means adding one adapter; the engine does not change.

The engine talks to one port; adapters absorb each agent difference



Kru provider layers · engine, port, 3 adapters, transports, agent processes

Arrows run both ways: commands go down, events and tool-call requests come up. Dashed boxes are separate processes that Kru starts and can lose at any time.

What each layer owns

Layer	Owns	Knows about	Milestone
Engine	What runs and when: queue, lanes, tool router, SQLite	The protocol only	M2–M7

Layer	Owns	Knows about	Milestone
`ProviderClient` port	The contract: protocol methods and declared capabilities	Nothing below it	M1
Adapters	Translation between the protocol and one agent's native API	One agent	M1
Transport	Framed messages over stdio, request ids, timeouts	Bytes and JSON-RPC	M0
Agent process	The model, its tools, its own session files	Nothing about Kru	External

The shared language. OpenBot uses the Codex App Server protocol as the language every adapter speaks, because Codex already had a clean JSON-RPC design. Kru keeps the same method names (see Core types) but gives each method's params and result a TypeScript type, so the compiler checks every adapter. JSON-RPC framing stays inside the transport.

How each adapter translates

Kru protocol	Codex adapter	Claude adapter	ACP adapter
`thread/start`	Sent as-is	Creates an SDK session itself	`session/new`
`turn/start`	Sent as-is	Calls the SDK's `query()` with the prompt	`session/prompt`
`item/agentMessage/delta` (up)	Received as-is	Built from SDK assistant messages	Built from `session/update` (`agent_message_chunk`)
`turn/completed` (up)	Received as-is	Built when the SDK result message arrives	Built when `session/prompt` returns

Kru protocol	Codex adapter	Claude adapter	ACP adapter
`item/tool/call` (up, request)	Received as-is	Built from the SDK tool callback	Built from ACP tool requests

The Codex adapter is thin. The Claude and ACP adapters are thick — about 2,400 and 3,150 lines in OpenBot — because they act as a JSON-RPC server: they answer requests themselves and build notifications from what their agent emits.

One turn through the layers (Claude)

1. The engine calls `turn/start` with the thread id and the prompt.
2. The Claude adapter calls `query()` and answers at once with the new turn id.
3. The SDK writes to the `claude` CLI's stdin and reads its stdout as stream-json.
4. Text arrives; the adapter emits `item/agentMessage/delta` notifications up to the engine.
5. The model calls `send_message`; the adapter sends an `item/tool/call` request up and waits for the engine's answer.
6. The SDK reports the result; the adapter emits `turn/completed`.

The engine sees the same six events it would see from Codex.

Where state lives

Layer	Keeps	When it is lost
Transport	Pending requests by id	Rejected on process exit (M0)
Adapter	Native session ids, the active turn	Session resumed or replaced with a handoff (M6)
Engine	Threads, deliveries, messages in SQLite	Never: SQLite is the source of truth

Rules that keep the layers clean

1. Dependencies point inward: the engine imports the port, never an adapter. Adapters are chosen in one registry.
2. An adapter never writes to SQLite or to the screen. It only emits protocol events.

3. The transport knows nothing about turns, threads or tools.
4. Each adapter declares a `capabilities` object (compaction mode, steering, subagents). The engine uses a feature when it is present, instead of reducing every agent to the lowest common denominator.
5. Every adapter passes the same conformance test suite: the fake, Codex and Claude run the same file.

Why it pays off

- **One file per agent.** A new provider is a new adapter; nothing else changes.
- **Free tests.** The fake agent is just another adapter, so M2–M7 run with no model and no API key.
- **Contained failures.** A crashed CLI becomes one `exit` event; the engine marks the turn interrupted and nothing above needs to know why.
- **Provider switching.** The thread never belonged to Codex or Claude, so it can move between them (M6).
- **Comparable results.** Every agent looks the same to the engine, so the same task can run on each and be measured — the base for evals and cross-provider review after v0.1.

Stack, ground rules and layout

Use TypeScript on Bun with `bun:sqlite`, the same stack as OpenBot, so you can compare files side by side. Use plain `async` / `await`; learn Effect later as a separate project.

Ground rules

1. Build the fake provider first. Every test runs against it: deterministic, free, and able to stall or crash on command. OpenBot does the same in `scripts/provider-fakes/`.
2. Write your version before you read theirs. Then read their file and note what they handled that you missed.
3. Every milestone ends with a test that proves its "Done when" line, and a small user-facing result.
4. SQLite is the source of truth. Provider session files are cache.
5. Never retry a turn whose outcome is unknown. Mark it `interrupted` and let a person decide.

Repository layout

```
kru/
  src/
    rpc/          JSON-RPC over stdio (M0)
    providers/    ProviderClient + adapters: fake, claude, codex (M1)
    store/        SQLite, events, projections, migrations (M2, M4)
    mailbox/      enqueue, delivery state machine (M2)
    drain/        lanes and scheduler (M3)
    turns/        notification handling, delta buffer (M4)
    tools/        tool router, send_message, history tools (M5, M6)
    threads/      provider sessions, handoff (M6)
    boot/         crash recovery (M7)
    api/          HTTP + SSE server (v1.0)
    cli/          kru command (every milestone)
  test/
    fakes/        scripted fake agents: echo, stall, crash, tool-caller
  examples/
  docs/
```

Core types

Five types carry the whole engine. Start with these and change them as the milestones teach you more.

```
// Who sent a delivery. Users, agents and routines share one queue.
type Sender =
  | { kind: "user" }
  | { kind: "agent"; agentId: string }
  | { kind: "routine"; routineId: string };

type DeliveryStatus = "queued" | "starting" | "running" | "completed" |
"failed" | "interrupted" | "cancelled";

// One unit of work for one agent. A row in SQLite, never only in memory.
interface Delivery {
  id: string;
  recipientAgentId: string;
  laneId: string | null;          // null = the agent's main chat
  sender: Sender;
  text: string;
  replyToMessageId: string | null;
  expectsReply: boolean;
```

```
  status: DeliveryStatus;
  idempotencyKey: string | null;  // UNIQUE: a retry returns the first
receipt
  createdAt: string;
}

// Kru's thread. Survives provider switches.
interface Thread { id: string; agentId: string; laneId: string | null }

// The provider's private session, bound to one thread. Disposable.
interface ProviderSession { threadId: string; provider: string;
externalSessionId: string; retiredAt: string | null }

// Every provider adapter implements this. One protocol for all of them.
interface ProviderClient {
  readonly provider: string;
  start(): Promise<void>;
  stop(): Promise<void>;
  request<T>(method: string, params: unknown, timeoutMs?: number):
Promise<T>;
  respond(id: string | number, result: unknown): void;
  respondError(id: string | number, error: { code: number; message: string
}): void;
  on(event: "notification", fn: (n: { method: string; params: any }) =>
void): void;
  on(event: "request", fn: (r: { id: string | number; method: string; params:
any }) => void): void;
  on(event: "exit", fn: (error: Error) => void): void;
}
```

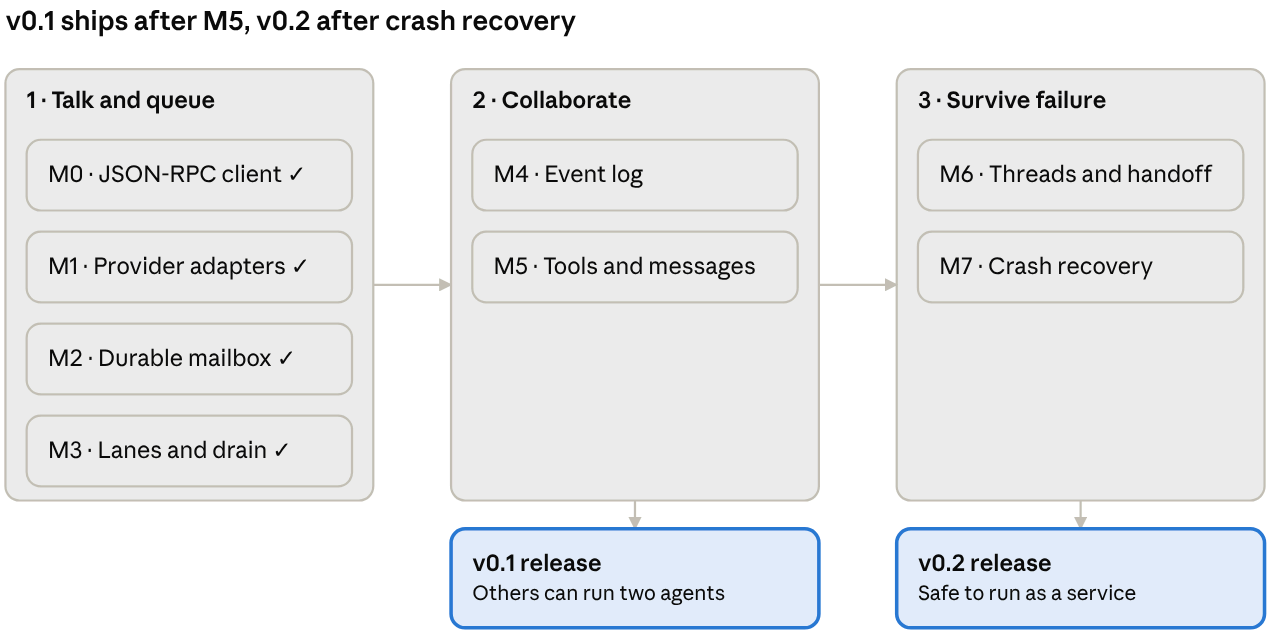
The protocol methods every adapter must speak:

Direction	Method	Meaning
Kru → provider	`thread/start`, `thread/resume`	Open or reopen a native session
Kru → provider	`turn/start`, `turn/interrupt`	Run one turn, or stop it
Provider → Kru	`turn/started`, `turn/completed`	Turn boundaries, with status

Direction	Method	Meaning
Provider → Kru	`item/started`, `item/agentMessage/delta`, `item/completed`	Streamed output and tool steps
Provider → Kru	`thread/tokenUsage/updated`	Context fill, for compaction
Provider → Kru (request)	`item/tool/call`	The model calls a Kru tool and waits for the answer

Milestones

Eight milestones, each isolating one idea. After M5 Kru is usable by other people (v0.1). After M7 it is crash-safe (v0.2). Each milestone lists what to build, which OpenBot files to read after you build it, the test that proves it, and what a user can do with it.



Kru roadmap · 8 milestones in 3 phases, 2 releases

Each phase depends on the one before it; inside a phase, build in order.

M0 — JSON-RPC over stdio (the transport) · Done

Status: done. Released as v0.0.1 on 2026-11-02, with the review fixes in v0.0.2 the same day. The spec below is what was built; the implementation record at the end of this milestone says how.

Purpose. Kru does not run a model. It starts agent CLIs as child processes and talks to them over stdin and stdout. M0 builds the bottom layer of the hexagon: a generic connection that sends and receives JSON-RPC messages reliably. It knows nothing about Codex, Claude, turns or threads, and it is proven against a fake agent. In M1 the Codex and ACP adapters use it directly; the Claude adapter uses the Agent SDK instead.

Public API

```
interface RpcConnectionOptions {
  command: string;
  args: string[];
  env?: Record<string, string>;
  requestTimeoutMs: number;
  maxLineBytes: number;        // one message = one line; longer = fatal
  maxQueuedWriteBytes: number; // stdin backpressure cap
  maxBadLines: number;         // consecutive unparsable lines before failing
}

class RpcConnection {
  constructor(options: RpcConnectionOptions);
  start(): void;
  stop(): Promise<void>;       // ends the whole process tree
  request<T>(method: string, params: unknown,
    options?: { timeoutMs?: number; signal?: AbortSignal; decode?: (value:
unknown) => T }): Promise<T>;
  notify(method: string, params?: unknown): Promise<void>;
  // The transport sends the answer: the resolved value, or an error if it
throws.
  onRequest(handler: (request: { id: string | number; method: string; params:
unknown; signal: AbortSignal }) => Promise<unknown>): void;
  on(event: "notification", listener: (n: { method: string; params: unknown
}) => void): void;
  on(event: "diagnostic", listener: (d: Diagnostic) => void): void; // stderr
| bad-line | late-response | write-blocked
  on(event: "exit", listener: (error: ProcessExitError) => void): void; //
fires exactly once
}
```

Design rules

1. **Pure transport.** No method names, turns or threads inside this module. Agent-specific behaviour, such as a liveness probe that needs a cheap method, belongs to the M1 adapter.
2. **Frame on bytes.** Split on the newline byte before decoding UTF-8, so a character split across two chunks is never corrupted.
3. **Every request gets exactly one answer.** The handler's value becomes the result; a thrown error becomes `-32603`; no handler becomes `-32601`. A deadlock from a missing listener is impossible.
4. **A bad line is a diagnostic, not a crash.** Messages before and after it in the same chunk are still handled. After `maxBadLines` consecutive bad lines the connection fails, so a truly broken CLI is still restarted.
5. **Backpressure.** When `stdin.write()` returns false, wait for `drain`. Queued bytes above `maxQueuedWriteBytes` reject the write with a clear error instead of growing memory.
6. **Late answers are visible.** Remember recent timed-out ids in a bounded set. An answer that arrives later becomes a `late-response` diagnostic with the method and the delay.
7. **Stale-process guard.** Events from an old child are ignored after a restart, and request ids never reset, so an old answer cannot resolve a new request.
8. **Exit reports the real reason.** On exit, wait briefly for stderr to end, then reject every pending request with a `ProcessExitError` that carries the last stderr line.
9. **Stop ends the whole tree.** Terminate the child and its children; force-kill after a grace period, so no MCP servers are left behind.
10. **One line has a size limit.** Past `maxLineBytes` the stream cannot recover, so the connection fails. OpenBot sets 256 MiB, because one Codex thread resume reached 21 MB.

Fake agent (`test/fakes/fake-agent.ts`) — a script that speaks the protocol and, on command: answers normally, answers out of order, stalls, crashes, sends notifications, sends a request back, writes split, merged or invalid lines, writes to stderr, and stops reading stdin. Every later milestone reuses it.

Tests — `bun test`, no network, no API key, about 2 seconds.

#	Test	Proves
1	`ping` returns `pong`	Basic round trip
2	100 concurrent requests answered in random order	Each answer reaches its own caller
3	Error response	Rejects with the code and message
4	Fake stalls	Rejects with a timeout error
5	Five notifications	Arrive in order
6	Fake sends a request; the handler answers	The fake receives the answer and echoes it back
7	One message split across two chunks; two messages in one chunk	Framing is correct
8	A bad line between two good lines in one chunk	Both good messages handled; one `bad-line` diagnostic; connection stays up
9	Fake exits during a request	Every pending request rejects; `exit` fires once, with the last stderr line
10	`stop()`, then `request()`	Child tree is gone; the new request rejects at once
11	Fake writes to stderr	Reported as a diagnostic, never parsed as protocol
12	Fake stops reading; 50 MB of writes	Memory stays bounded; writes wait or fail cleanly
13	Timeout, then the fake answers late	One `late-response` diagnostic; no crash, no double resolve
14	Fake sends a request with no handler set	It receives a `-32601` error

After every test: the pending map is empty, no child process is running, and there are no unhandled promise rejections.

Done when: all 14 tests and the hygiene checks pass, every time.

You can now:

```
$ kru rpc ping --fake
pong (1.4 ms)

$ kru rpc ping --fake --stall
error: request "ping" timed out after 2000 ms
```

Stretch: send `initialize` to a real `codex app-server --listen stdio://` and print the answer.

Read after: `src/backend/app-server-client.ts`, `src/backend/jsonl.ts`, `src/backend/protocol.ts`.

Out of scope: providers, SQLite, queues, retries.

Improvements over OpenBot

OpenBot's transport is production-hardened. Kru keeps its byte framing, line limit, stale-process guard, stderr-before-exit and process-tree kill, and changes six things:

Area	OpenBot	Kru M0
Separation	The Codex client mixes transport with Codex logic (turn interrupts, resume params, history paging); ACP and Pi have their own transports	One generic transport for every JSON-RPC agent; agent logic lives in adapters
Bad line	Any unparsable line ends the process and every agent on it; valid messages earlier in the chunk are dropped	Diagnostic and skip; fail only after `maxBadLines` in a row
Backpressure	`stdin.write()` result ignored; writes queue without limit	Wait for `drain`; cap queued bytes
Late answers	Ignored silently after a timeout	Reported as `late-response`, so the engine can reconcile a turn that did start
Unanswered requests	The tool router answers unknown methods, but the transport does not guarantee an answer	The transport always answers exactly once

Area	OpenBot	Kru M0
Types	`params: unknown`, checked deep in the engine	Decoded at the boundary through `decode`

One trade-off: OpenBot's fail-fast on a bad line is a deliberate safety choice. Skipping is safe here only because each NDJSON line is independent, which is why the `maxBadLines` limit stays.

Implementation record

The M0 transport meets every design rule above and all six improvements over OpenBot. Code: aminhaiqal/kru; exports in `src/rpc/index.ts`, tests in `test/rpc.test.ts`.

Shipped in v0.0.1

- Generic JSON-RPC connection: byte framing, strict UTF-8, CRLF accepted, incoming and outgoing line limits, concurrent requests, notifications, inbound request handlers.
- Timeouts, cancellation and result decoding; bounded stdin writes with `drain`; a `late-response` diagnostic from a bounded set of the last 1,024 timed-out ids.
- Process-tree cleanup: each child in its own process group on POSIX, SIGTERM then SIGKILL; `taskkill /T /F` on Windows.
- The deterministic fake agent, `kru rpc ping --fake` with stall mode, and `kru --version`.
- 32 tests and GitHub Actions CI.

Beyond the plan

- An oversized outgoing frame rejects only its own write, not the connection.
- Cancelling or timing out a queued request removes its unsent frame.
- Handler replies from a previous process are suppressed after a restart.
- Each spawned process owns its own listeners, maps, buffers and handler controllers.
- Read-only `pid`, `pendingRequestCount` and `queuedWriteBytes` for observability.

Review fixes (v0.0.2)

Found in review	Fix
Codex omits the `jsonrpc` field; a strict check would turn every Codex message into a bad line	Accept a missing field, reject a wrong version, and set outgoing frames with the `jsonrpcField` option (default `true`); a Codex connection factory passes `false`
A 100 ms SIGTERM grace can corrupt a real agent's session files	Configurable shutdown grace, default 3 s; configurable stderr wait, default 500 ms; tests keep shorter values
Children in their own process group no longer receive Ctrl+C	Every registered connection stops on SIGINT or SIGTERM; handlers stay until cleanup ends; the CLI returns conventional signal exit codes
A hung shutdown makes the user wait out the grace period	A second signal sends SIGKILL to every tracked process group at once, skips the remaining waits, reaps the children and exits
After `kill -9` of Kru, children run on as orphans	Each child's PID and process group are saved with its start identity, for M7 recovery, and removed after cleanup
A crash during a write could leave a half-written ownership record	Records are published with an fsynced temporary file and an atomic rename; the reader ignores unpublished, truncated and invalid records
macOS can refuse a process-group signal during reaping	The transient permission error is handled

Verification

- Strict TypeScript checking passes.
- All 32 v0.0.1 tests pass in about 1.2 s with Bun 1.4.4 on macOS; ten repeated runs pass all 320 cases with the per-test hygiene checks.
- Ten repeated suite runs in Linux CI, with manual workflow dispatch.
- An exact-byte `initialize` smoke test passes against real Codex 0.165.0, locally and in CI.
- v0.0.2 tests cover envelope compatibility, graceful stop, the second-signal escalation, process ownership, the ownership reader and the signal lifecycle.

Open items

- [x] ~~Tag v0.0.2 before M1 starts~~
- [x] ~~A second Ctrl+C force-kills every tracked process group and exits~~
- [x] ~~Write the ownership file atomically; ignore truncated and invalid records~~
- [x] ~~Real Codex `initialize` smoke test with `jsonrpcField: false`~~
- [ ] Confirm the `late-response` diagnostic names the method as well as `delayMs`
- [ ] Validate process-tree cleanup on Windows

M1 — One protocol, many providers (the port and adapters) · Done

Status: done on 2026-11-03. Fake, Codex and Claude adapters run behind one typed port and pass the same conformance suite. Five follow-ups and the v0.0.3 tag are listed at the end of the implementation record, below the improvements table.

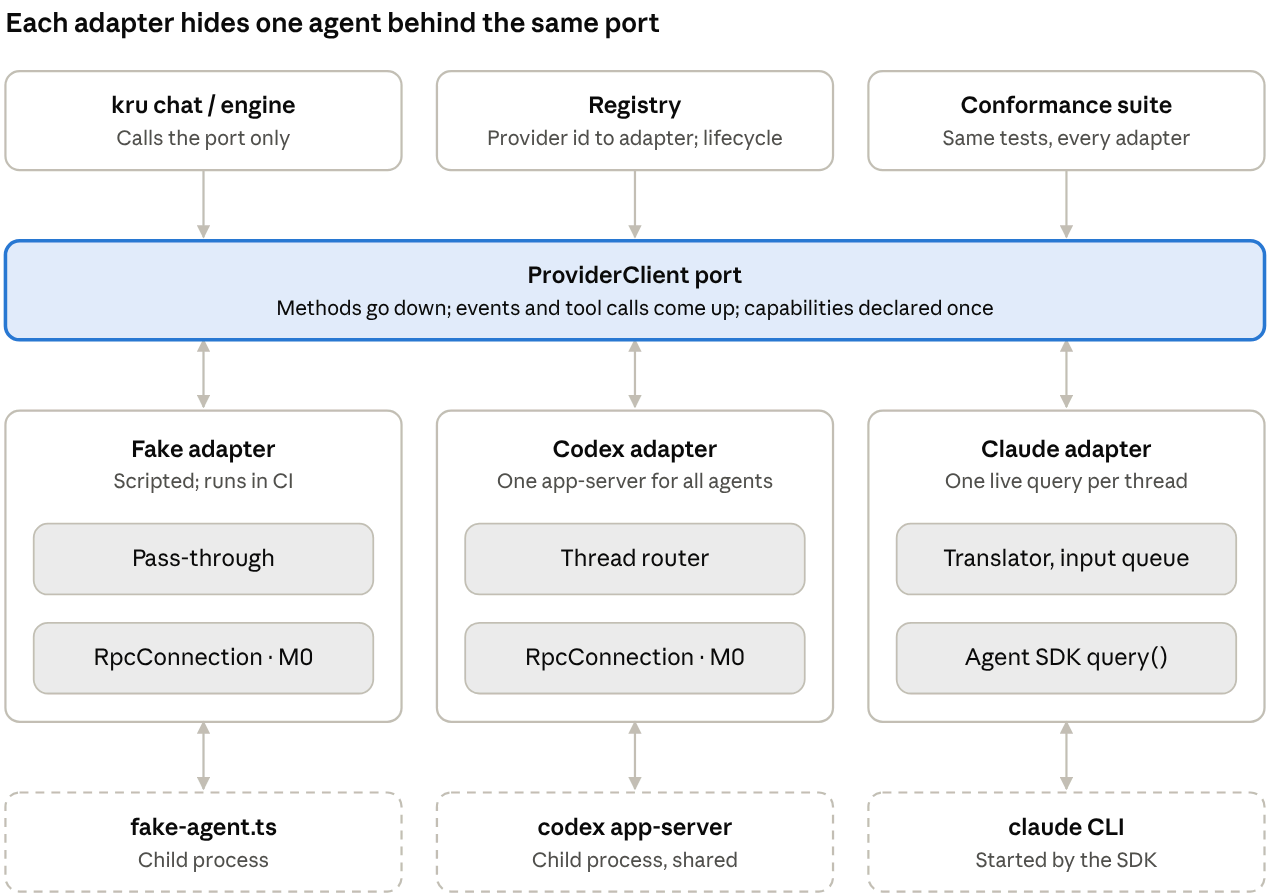
Purpose. M1 builds the `ProviderClient` port and the adapters behind it. After M1 the engine runs a turn on the fake, Codex or Claude without knowing which. M1 is also where token use becomes visible: every adapter must report what each turn cost, because the main reason people leave OpenBot is how many tokens it uses.

Build order

1. Typed protocol: one TypeScript map of methods (params and result) and one of events. The method names stay those in Core types.
2. A `capabilities` object that every adapter declares.
3. The `ProviderClient` port and a registry that picks the adapter.
4. The fake adapter on the M0 fake agent, with scripted token counts.
5. The conformance suite, written against the fake first.
6. The Codex adapter on `RpcConnection` through the existing Codex factory (`jsonrpcField: false`). One app-server process serves every Codex agent as separate threads.
7. The Claude adapter around the Agent SDK. It is the thick one, so it comes last.
8. The `kru chat` demo, which prints the usage of each turn.

Architecture

M1 has three parts: the port (the contract), the adapters (translation), and the registry with the conformance suite (wiring and proof).



M1 architecture · port, registry, conformance suite, 3 adapters, 3 processes

Commands go down; events and tool calls come up. Fake and Codex reuse the M0 transport; Claude goes through the Agent SDK.

The port carries four kinds of traffic

- **Down, methods:** `thread/start`, `thread/resume`, `turn/start`, `turn/interrupt`.
- **Up, events:** `turn/started`, message deltas, `item/completed`, usage, `turn/completed`.
- **Up, one request:** a tool call, which the engine's handler must answer.
- **Declared once:** `capabilities`.

Inside an adapter

Every adapter has the same four jobs:

1. Translate methods down into the agent's native calls.
2. Translate native output up into protocol events.

3. Bridge tool calls: the native tool call goes to `onToolCall(call, signal)`, and the result goes back in native form.
4. Normalise usage and errors.

An adapter keeps two pieces of state only: Kru's thread id mapped to the native session, and the active turn of each thread. A turn moves from `starting` to `running` to `completed`; a process exit from any state ends it `interrupted`.

How the three adapters differ

	Fake	Codex	Claude
Transport	M0 `RpcConnection`	M0 `RpcConnection`, `jsonrpcField: false`	Agent SDK in process; it starts the `claude` CLI
Processes	One per test	One app-server for every Codex agent	One live `query()` per active thread, closed when idle
Start / resume	Passed through	`thread/start` with instructions and tools / `thread/resume`	`query({ sessionId })` / `query({ resume })`
One turn	`turn/start`	`turn/start`	A message pushed into the thread's input queue; the turn ends at the SDK result message
Text streaming	Passed through	`item/agentMessage/delta`	SDK stream events (`includePartialMessages: true`)
Kru tools	Scripted	Declared at thread start; Codex sends `item/tool/call`	An in-process MCP server (`createSdkMcpServer`); the model sees `mcp__kru__send_message`
Usage source	Scripted	`thread/tokenUsage/updated`, last-turn value	SDK result message: input, cache read, cache write, output
Size	Trivial	Thin	Thick

Claude: streaming input

The Claude adapter does not start a new `query()` for each turn. It starts one per thread whose prompt is an async queue, and each `turn/start` pushes a message into it. The session stays alive between turns: no process restart, and the prompt cache stays warm. OpenBot does the same with `AsyncMessageQueue` in `src/backend/claude-client.ts`.

The tool-call path

1. The model calls a Kru tool; the adapter receives the native call and calls `onToolCall(call, signal)`.
2. The engine's handler does the work — in M5, it enqueues a delivery — and returns a result.
3. The adapter sends the result back in native form, and the model continues.

As in M0, every tool call gets exactly one answer, and `signal` aborts when the turn is interrupted or the process exits.

Registry and lifecycle

- `registry.create(provider, options)` returns a `ProviderClient`. Nothing else in Kru constructs an adapter.
- Adapters start on first use. The registry hands out one shared Codex instance.
- On a signal, the registry stops every adapter through the M0 signal handling.

The conformance suite (below) runs against `registry.create("fake" | "codex" | "claude")`. It is the executable definition of the port: an adapter that passes it works with the engine.

Folder layout

```
src/providers/
  protocol.ts          typed method and event maps, capabilities (depends on
nothing)
  port.ts              ProviderClient interface
  registry.ts          provider id → adapter factory, lifecycle
  fake/adapter.ts
  codex/adapter.ts     uses src/rpc (M0)
  claude/adapter.ts    uses @anthropic-ai/claude-agent-sdk
  claude/input-queue.ts
test/providers/conformance.test.ts
```

Dependencies point inward: adapters import `protocol.ts` and their own transport; the engine imports `port.ts` and `registry.ts` only.

Port API

```
interface ProtocolMethods {
  "thread/start":   { params: { cwd: string; instructions: string; tools:
ToolDefinition[]; model: string;
                      promptProfile: "minimal" | "coding" | "full";
                      permissions: { sandbox: "read-only" | "workspace-
```

```
write"; approvalPolicy: "on-request" | "never" } }; result: { threadId:
string } };
  "thread/resume":  { params: { threadId: string }; result: { threadId:
string } };
  "turn/start":     { params: { threadId: string; input: string; model:
string }; result: { turnId: string } };
  "turn/interrupt": { params: { threadId: string; turnId: string }; result:
Record<string, never> };
}

interface ProtocolEvents {
  "turn/started":              { threadId: string; turnId: string };
  "item/agentMessage/delta":   { threadId: string; turnId: string; itemId:
string; delta: string };
  "item/completed":            { threadId: string; turnId: string; item:
ThreadItem };
  "thread/tokenUsage/updated": { threadId: string; turnId: string;
inputTokens: number; cachedInputTokens: number;
                                outputTokens: number; contextWindow: number
| null; estimated: boolean };
  "turn/completed":            { threadId: string; turnId: string; status:
"completed" | "failed" | "interrupted"; error?: string };
}

interface ProviderCapabilities {
  compaction: "events" | "request" | "native" | "handoff" | "none";
  steer: boolean;               // can add input to a running turn
  multiThreadProcess: boolean;  // one process serves many threads
  usage: "exact" | "estimated";
  promptCache: boolean;
}

interface ProviderClient {
  readonly provider: string;
  readonly capabilities: ProviderCapabilities;
  start(): Promise<void>;
  stop(): Promise<void>;
  request<M extends keyof ProtocolMethods>(method: M, params:
ProtocolMethods[M]["params"]): Promise<ProtocolMethods[M]["result"]>;
  on<E extends keyof ProtocolEvents>(event: E, listener: (params:
ProtocolEvents[E]) => void): void;
```

```
  onToolCall(handler: (call: ToolCall, signal: AbortSignal) =>
Promise<ToolResult>): void;
  onExit(listener: (error: Error) => void): void;
}
```

Design rules

1. The engine imports the port and the registry only, never an adapter.
2. An adapter never writes to SQLite or the screen. It only emits protocol events.
3. A turn that ends normally ends with exactly one `turn/completed`. If the process dies, the adapter emits `exit` and does not invent a completion: it does not know how the turn ended. The waiting turn rejects with an unknown outcome, and the engine decides what it was (M7).
4. Every turn reports usage. When the provider gives counts, `estimated` is false; when it does not, the adapter estimates (about 4 bytes a token) and sets `estimated: true`. A turn with no usage event fails conformance.
5. A stable prefix. Instructions and tool definitions are fixed when the thread starts and never injected per turn, so the provider's prompt cache stays valid. Changing them starts a new session, and the reason is logged.
6. A small surface. Kru's own instructions stay under 8 KB (about 2,000 tokens) and an agent sees at most 6 tools in M1. A test fails when either grows, as OpenBot's `check:agent-context` does, but with a lower limit.
7. Shared processes. When `multiThreadProcess` is true (Codex), one process serves all agents of that provider. Claude threads with no turn for 10 minutes are closed and resumed on demand.
8. Capabilities, not the lowest common denominator. The engine uses a feature when an adapter declares it.
9. A prompt profile per agent. `minimal` is a short custom prefix with no built-in tools. `coding` is a short coding prefix with read tools for read-only agents or coding tools for writable ones. `full` keeps the agent CLI's own system prompt, project settings and standard tools, and is the baseline for measuring the saving (a follow-up). For Claude, `minimal` and `coding` use a custom `systemPrompt`, `settingSources: []` and an explicit tool list.

Conformance suite — one test file; every adapter runs it.

#	Test	Proves
1	Start a thread, run a turn	Deltas arrive, then one `turn/completed`
2	Interrupt during a turn	Status `interrupted`; still exactly one `turn/completed`
3	The model calls an `echo` tool	The handler's result reaches the model
4	The process exits during a turn	`turn/completed` with `interrupted`, then `exit`
5	Restart the adapter and resume the thread	The next turn works on the same thread
6	Usage	One or more usage events per turn; counts are not negative; `estimated` matches `capabilities.usage`
7	Prefix stability	Two turns on one thread send identical instructions and tool definitions
8	Context budget	Instructions under 8 KB; 6 tools or fewer

The fake runs the suite in CI on every push. Codex and Claude run it in a nightly canary on the cheapest model, because it needs keys and costs tokens.

Watch for: providers that stream text before they announce the item — create the message on the first delta. Codex reports usage as both a last-turn and a running total; take the last-turn value, never add the totals.

Done when: `kru chat` works the same with `--provider fake`, `codex` and `claude`; the conformance suite passes for all three; and every turn prints its usage.

You can now (example output):

```
$ kru chat --provider claude
> summarise this repository
...
[turn 1 · input 12,410 tokens (10,880 cached) · output 642]
```

Read after: `src/backend/agent-client.ts`, `src/backend/claude-client.ts` (the method switch near line 494 and the functions that emit `item/*`), `src/backend/acp-client.ts`, `src/backend/provider-drivers.ts`, `src/backend/agent/usage-`

`collection.ts`, `scripts/provider-fakes/agents/conformance.ts`, `tools/agent-context/budget.json`.

Improvements over OpenBot

Area	OpenBot	Kru M1
Types	String methods; `params: unknown` read with `getString`	Typed method and event maps; the compiler checks each adapter
Capabilities	Optional fields spread over the client	One declared object
Usage	Collected from provider-specific notifications in the turn lifecycle	A required event; every adapter emits it, exact or estimated
Added context	About 14 KB of instructions and 14 KB of visible tools — roughly 7,000 tokens on every model step, before memories and teammates	Under 8 KB of instructions and at most 6 tools, enforced by a test
Per-turn text	ACP and Muse insert instructions on each normal turn	Fixed at thread start, never per turn
Claude system prompt	Always the full `claude_code` preset with OpenBot's instructions appended	`minimal` or `coding` per agent, with `full` as the baseline; the saving is measured
Conformance	A fake agent per scenario	One suite that every adapter must pass

Implementation record

Code: aminhaiqal/kru; port in `src/providers/protocol.ts`, adapters in `src/providers/{fake,codex,claude}/`, turn consumer in `src/chat/turn.ts`.

Delivered

- **Typed port.** Four methods with checked params and results. Typed notifications for turn boundaries, items, deltas, usage, tool calls and `approval/request`. Every tool or approval request gets exactly one reply, with a deadline and an abort signal. Request ids are never reused, even after a restart.

- **Fake adapter.** `echo`, `stall`, `crash`, `failure` and `tool` modes; deterministic usage values for tests.
- **Codex adapter.** One shared `CodexRuntime` (one app-server) for every agent, with thread ownership that routes events, tool calls and approvals to the right agent. Stopping one agent leaves the others running; the last one stops the process. Uses the experimental dynamic-tool API and `jsonrpcField: false`.
- **Claude adapter.** A custom `systemPrompt`, `settingSources: []` and an explicit built-in tool list. An in-process MCP server for Kru's tools. One persistent streaming input queue per thread. Invented turn ids. Per-turn `usage` kept apart from the session's cumulative `modelUsage`. Resume through the SDK's session metadata.
- **Agent configuration.** `cwd`, `model`, `instructions`, `tools`, `promptProfile` and `permissions` (default `read-only` and `on-request`), also as CLI flags.
- **Prefix control.** Stable profile text, canonical tool order and a snapshot at thread start. `Thread.prompt` reports the profile, instruction bytes, tool count and a SHA-256 prefix hash. Instructions of 8,000 bytes or more and a seventh tool are refused at start and at resume.
- **Approvals.** A missing handler declines; nothing hangs and nothing is granted by default. The CLI declines every request.
- **Fault isolation.** A malformed known notification fails only its turn; three in a row end the process.
- **Chat.** An idle timeout (default 120,000 ms) that resets on activity from the same thread. Usage on stderr after each turn, or `usage: unavailable`.
- **Shutdown.** M0's signal handling for all providers. Claude subprocesses also run in process groups with ownership records.

Changes from the plan

Plan	Built	Why
Rule 3: the adapter emits `turn/completed` with `interrupted` when the process dies	`exit` only; the turn's outcome stays unknown	The adapter cannot know how the turn ended. Rule 3 now says so.
`promptProfile`: `full` or `lean`	`minimal` and `coding`, both short custom prefixes	`full` is a follow-up, needed as the measuring baseline

Plan	Built	Why
At most 6 Kru tools	For Claude, at most 6 built-in and Kru tools together	Leaves no room for M5's tools; a follow-up separates the budgets
A wall-clock turn timeout	An idle timeout	A turn that keeps making progress can run as long as it needs

Verification

- The same conformance cases, including #7 (prefix stability) and #8 (budget), run against all three adapters. Codex runs against a scripted stdio peer; Claude against SDK events and a real in-memory MCP transport.
- Ten repeated offline runs: 1,020 passes, no failures. Strict TypeScript and a frozen install pass.
- Opt-in live tests pass on real Codex and Claude: streaming, usage, resume, tool calls and interruption through the common port.
- `codex-canary.yml` runs nightly at 03:17 UTC against the latest Codex. It checks the `initialize` bytes, dynamic-tool registration and the generated protocol schemas, with no model call. Validated locally against Codex 0.166.0; pinned CI stays on 0.165.0.

Profile benchmark (2026-11-03)

One ordered pair, `full` first, on Claude Sonnet 4.6 (`claude-sonnet-4-6`): repair a TypeScript duration parser with compound h/m/s/ms parts, strict ordering, full-input validation and safe-integer overflow checks. Both profiles passed all 26 independent acceptance checks with unchanged tests.

Measurement	Full	Coding
Input tokens, total (includes cache reads and writes)	161,579	98,257
Cache reads	147,055	66,954
Cache writes	14,515	31,295
Output tokens	8,697	13,896
Input plus output	170,276	112,153 (−34.1%)

Measurement	Full	Coding
Elapsed time	110.7 s	178.8 s (+61.4%)
Cost at list prices, 5-minute cache writes	$0.229	$0.346 (+51%)

Cost uses Sonnet 4.6 list prices per million tokens: $3 input, $15 output, $3.75 for 5-minute cache writes and $0.30 for cache reads (Claude docs). With 1-hour cache writes, `coding` costs 59% more.

Finding: fewer tokens did not mean lower cost. Almost all of the full profile's input was cache reads, the cheapest tokens. The lean profile cut those, but produced more output and twice the cache writes, and took longer. A likely reason, still to test: Claude Code's full system prompt guides the model to work in fewer steps. `full` stays the default profile for coding agents until a controlled benchmark shows otherwise.

Limits: one ordered pair; uncontrolled caching; system prompt, tool list and setting sources changed together. Dollar savings in general and quality parity on other tasks are not measured.

Next benchmark: at least 5 pairs in alternating order with the median and range; a controlled cache (a unique marker for cold starts, or a warm-up run); one change at a time (full prompt with restricted tools, lean prompt with full tools, the current lean profile); model steps and tool calls counted per turn; and a run on a current model.

Follow-ups

- [x] ~~Normalise usage across providers. Claude reports cache reads and writes apart from `input_tokens`; Codex counts cached tokens inside its input. Report `inputTotal`, `cachedRead`, `cacheWrite` and `output`; print the cached part in the CLI; add a conformance check that `inputTotal` is at least `cachedRead + cacheWrite`.~~
- [x] ~~Separate the tool budgets: at most 6 Kru-declared tools; built-in tools chosen by the profile, reported but not capped.~~
- [x] ~~Add the `full` profile (the `claude_code` preset and project settings), then compare `full` and `coding` on one task for tokens and result.~~
- [x] ~~Permission parity: `workspace-write` on Claude allows file tools inside `cwd` and its roots and asks only for shell commands or outside paths. Add a conformance case: the same settings allow and refuse the same actions on every adapter.~~
- [x] ~~Confirm that output or progress from a long shell command resets the idle timer on both adapters.~~

- [x] ~~Merge the canary to the default branch so its schedule runs.~~
- [x] ~~Tag v0.0.3.~~

M2 — The durable mailbox · Done

Status: done on 2026-11-04 and released as v0.0.4. The mailbox stores deliveries and moves them between states; it starts no provider. The implementation record and follow-ups are below the improvements table.

Purpose. M2 makes Kru durable. The mailbox is a SQLite inbox through which all work enters — from a person, another agent or a routine. M2 only stores work and moves it between states; it never runs a turn. Running is M3's job. Keeping the two apart is what lets M7 recover after a crash.

Architecture

- `src/store/db.ts` opens `bun:sqlite` with WAL mode, a busy timeout and foreign keys on, and runs migrations from version 1.
- `src/mailbox/` holds the mailbox API, the delivery state machine and the queue queries.
- `src/agents/` is a minimal agent table, so `enqueue` can check that a recipient exists.
- The CLI adds `kru agent create`, `kru send`, `kru queue` and `kru cancel`.

The mailbox is a port, as the provider is in M1. M4 replaces its internals with the event log without changing a caller.

Data model: messages apart from deliveries

```
CREATE TABLE agents (
  id          TEXT PRIMARY KEY,
  name        TEXT NOT NULL UNIQUE,
  provider    TEXT NOT NULL,
  config_json TEXT NOT NULL CHECK (json_valid(config_json)),
  created_at  TEXT NOT NULL
);

CREATE TABLE messages (
  id              TEXT PRIMARY KEY,
  sender_kind     TEXT NOT NULL CHECK (sender_kind IN
('user','agent','routine')),
  sender_agent_id TEXT REFERENCES agents(id),
  text            TEXT NOT NULL,
```

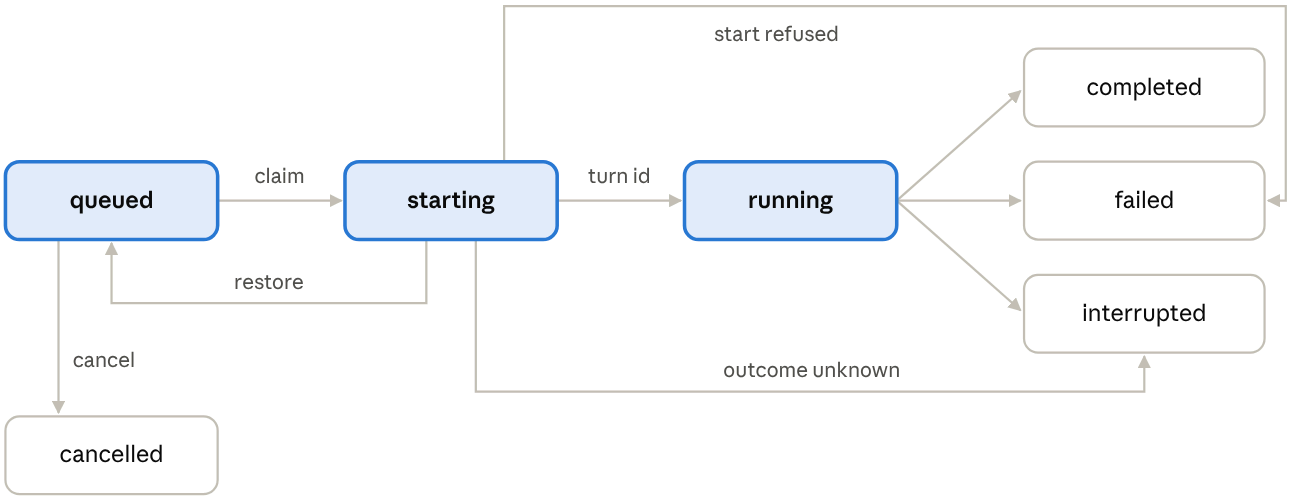
```
  reply_to        TEXT,
  idempotency_key TEXT UNIQUE,          -- NULL allowed; duplicates refused
  request_hash    TEXT,                 -- same key with other content =
client error
  created_at      TEXT NOT NULL
);

CREATE TABLE deliveries (
  seq          INTEGER PRIMARY KEY,     -- queue order; never order by time
  id           TEXT NOT NULL UNIQUE,
  message_id   TEXT NOT NULL REFERENCES messages(id),
  recipient_id TEXT NOT NULL REFERENCES agents(id),
  lane_id      TEXT,                    -- NULL = the agent's main chat
  status       TEXT NOT NULL,
  turn_id      TEXT,
  error        TEXT,
  retry_of     TEXT REFERENCES deliveries(id),
  chain_id     TEXT NOT NULL,           -- M5 hop limit; in the schema from
version 1
  hop          INTEGER NOT NULL DEFAULT 0,
  updated_at   TEXT NOT NULL
);
CREATE INDEX deliveries_queue ON deliveries (recipient_id, lane_id, status,
seq);
```

One message can go to several agents: `send_message` to B and C is one message with two deliveries, each with its own status. OpenBot splits them the same way (`projection_mailbox_messages` and `projection_deliveries`). The auto-increment `seq` gives a strict order; two messages can share a millisecond, and clocks can jump.

The delivery state machine

A delivery only moves forward; final states never reopen



Tinted: open states. Outlined: final states; a retry creates a new delivery.

Delivery state machine · 3 open states, 4 final states

A drain claims a queued delivery into `starting`; a confirmed turn id moves it to `running`. If `turn/start` fails before a turn begins, the delivery is restored to `queued` at its original place, or ends `failed` when the provider refused it. If a turn may have begun, it ends `interrupted` and is never retried automatically.

Mailbox API

```
type DeliveryStatus = "queued" | "starting" | "running" | "completed" |
"failed" | "interrupted" | "cancelled";

interface EnqueueInput {
  sender: Sender;
  recipientIds: string[];
  text: string;
  laneId?: string | null;
  replyTo?: string | null;
  idempotencyKey?: string;
  chain?: { id: string; hop: number };   // set by M5; otherwise a new chain
}

interface EnqueueReceipt { messageId: string; deliveryIds: string[];
duplicate: boolean }

interface Mailbox {
  enqueue(input: EnqueueInput): EnqueueReceipt;
```

```
  claimNext(agentId: string, laneId: string | null): Delivery | null; //
queued → starting
  markRunning(deliveryId: string, turnId: string): void;              //
starting → running
  finish(deliveryId: string, status: "completed" | "failed" | "interrupted",
error?: string): void;
  restore(deliveryId: string): void;                                  //
starting → queued
  cancel(deliveryId: string): void;                                   //
queued → cancelled
  retry(deliveryId: string): EnqueueReceipt;                          //
final → a new delivery
  get(deliveryId: string): Delivery | null;
  listQueue(agentId: string, laneId?: string | null): Delivery[];
  onChange(listener: (change: { deliveryId: string; status: DeliveryStatus })
=> void): () => void;
}
```

Design rules

1. Store and transition only. The mailbox never runs a turn.
2. Every transition is a compare-and-set update, checked against one table of allowed transitions: `UPDATE deliveries SET status = ? WHERE id = ? AND status = ?`. A change count other than 1 means another writer won. This is what lets M3 run several drains without claiming the same work twice.
3. A final state never reopens. A retry creates a new delivery with `retry_of`, so the history shows that the first attempt failed.
4. Idempotency in the same transaction as the insert: a `UNIQUE` key plus a hash of the request. A duplicate returns the first receipt with `duplicate: true`; the same key with other content is refused.
5. Notify after commit. `onChange` fires only after the transaction commits, so M3 never schedules work that was rolled back.
6. Queue order is `seq`, never a timestamp. `restore` keeps the delivery's `seq`, so it does not move to the back of the queue.
7. One writer at a time. WAL mode, `BEGIN IMMEDIATE` for every write and a busy timeout. The API is synchronous, as `bun:sqlite` is, so no mailbox operation can be interleaved halfway through.

8. Migrations from version 1. Each schema change and its `schema_migrations` row commit in one transaction. Every shipped version keeps a fixture, and a parity test checks that a new database and a migrated one have the same schema.
9. Fields for later milestones now: `chain_id` and `hop` for M5; `sender_kind` is the turn cause for M4.

Tests

#	Test	Proves
1	The same key sent twice	One message; the second call returns the first receipt with `duplicate: true`
2	The same key with other text	Refused; nothing written
3	One message to two agents	Two deliveries with independent statuses
4	Restart	Queued rows and their order survive
5	`completed → running`	Refused; the row does not change
6	Two connections call `claimNext` at the same time	They never claim the same delivery
7	`claimNext` on one lane	Takes the lowest `seq` of that lane; other lanes do not change
8	`restore`	Back to `queued` with the original `seq`
9	`cancel`	Works on `queued`; refused on `running`
10	`retry` of a final delivery	A new delivery with `retry_of`; the old one does not change
11	A transaction that rolls back	No `onChange` event
12	Migrations	An empty database and a version-1 fixture end with the same schema
13	An unknown recipient	Refused; nothing written

Done when: all 13 tests pass in ten repeated runs, as in M0 and M1.

Watch for: `UPDATE … RETURNING` needs SQLite 3.35 or later — check the version that Bun bundles. A CLI and the M3 daemon can write at the same time; without a busy timeout, one of them gets `SQLITE_BUSY`.

Optional bridge to M3: `kru deliver --once <agent>` claims one delivery, runs it with the M1 `runChatTurn` and records the outcome. It is the smallest path from message to turn to final state; M3 automates it.

You can now (example output):

```
$ kru agent create reviewer --provider claude
$ kru send reviewer "review the latest commit"
queued  message 7f3c…  delivery d-1
$ kru queue reviewer
d-1  queued  user  "review the latest commit"
```

Read after: `src/backend/mailbox-store.ts` (`enqueue`, `markStarting`, `restoreQueued`, `markTerminal`), the `projection_mailbox_messages` and `projection_deliveries` tables in `src/backend/openbot-database-schema.ts`, "Database migrations" in `src/backend/AGENTS.md`, and `src/backend/openbot-database-schema-parity.test.ts`.

Improvements over OpenBot

Area	OpenBot	Kru M2
Scope	`mailbox-store.ts` is about 3,300 lines: deliveries, attachments, reactions, edits and a file-deletion outbox	Deliveries only; attachments get their own module later
Idempotency	Keys in the store's in-memory state map, pruned by age	A `UNIQUE` column written in the same transaction, and a check that the content matches the key
Transitions	Status checks spread through the store	One allowed-transition table and compare-and-set updates
Retry	The same delivery is requeued	A new delivery that links to the old one
Chain and hop fields	Not part of the design	In the schema from version 1

Implementation record

Code: `src/store/` (opening, migrations, writer transactions), `src/store/migrations/001-initial.sql`, `src/agents/`, `src/mailbox/` (port in `types.ts`, transition table in `state.ts`, SQLite adapter in `sqlite.ts`) and `src/cli/mailbox.ts`.

Delivered

- **Store.** Foreign keys, WAL, a 5-second busy timeout and `synchronous=FULL`. SQLite 3.35 or later is checked at startup (verified on 3.51.0). Every write takes `BEGIN IMMEDIATE` before it reads the state it changes, and public writes refuse to run inside another transaction.
- **CLI.** `agent create` and `agent list`; `send` to one or more agents with `--lane`, `--main`, `--reply-to` and `--expects-reply`; `queue`, `cancel` and `retry`. Agents are named by id or unique name; queue output escapes message text.
- **Transitions.** One allowed-transition table; every change requires exactly one changed row. `seq` uses `AUTOINCREMENT`, so queue positions are never reused. A retry creates a new queued delivery with `retry_of`, keeping the recipient, lane, chain and hop.
- **Idempotency.** A unique key and a SHA-256 hash of the normalised request (sender, sorted recipients, text, lane, reply target, reply flag, chain) in the same transaction as all deliveries. A replay returns the original ids with `duplicate: true`, even after a restart or a retry. Another request under the same key throws `IdempotencyConflictError` and writes nothing. Duplicate recipients are refused.
- **Schema additions.** `sender_routine_id` and `expects_reply`, to match Core types; SQL checks on sender shape, statuses, JSON configuration and key/hash pairs.
- **Notifications.** `onChange` fires after commit, in commit order, never for replays or rolled-back writes. A listener error becomes a warning. Notifications are local hints; SQLite is the source of truth.
- **Migrations.** One migration path for new and existing databases, so no separate latest schema can drift. A frozen v1 fixture and a parity test check schema, retained data and foreign keys.

Changes from the plan

Plan	Built	Why
The diagram had no arrow from `starting` to `failed`	Follows the text, which allows it; the diagram is now fixed	A provider can refuse a start before a turn begins

Plan	Built	Why
A fresh schema plus migrations, checked by a parity test	One migration path for both	Removes the cause of drift instead of testing for it
The sample SQL	Adds `sender_routine_id` and `expects_reply`	Matches the core types
Optional `kru deliver --once`	Left for M3	M3 runs deliveries

Verification

- The numbered cases in `test/mailbox.test.ts` and the v1 parity test in `test/store.test.ts` cover all 13 tests.
- 24 M2 tests run ten times: 240 passes, no failures, including real competing Bun processes on one file.
- Full check: TypeScript passes; 162 tests pass, 9 opt-in tests skipped.

Follow-ups

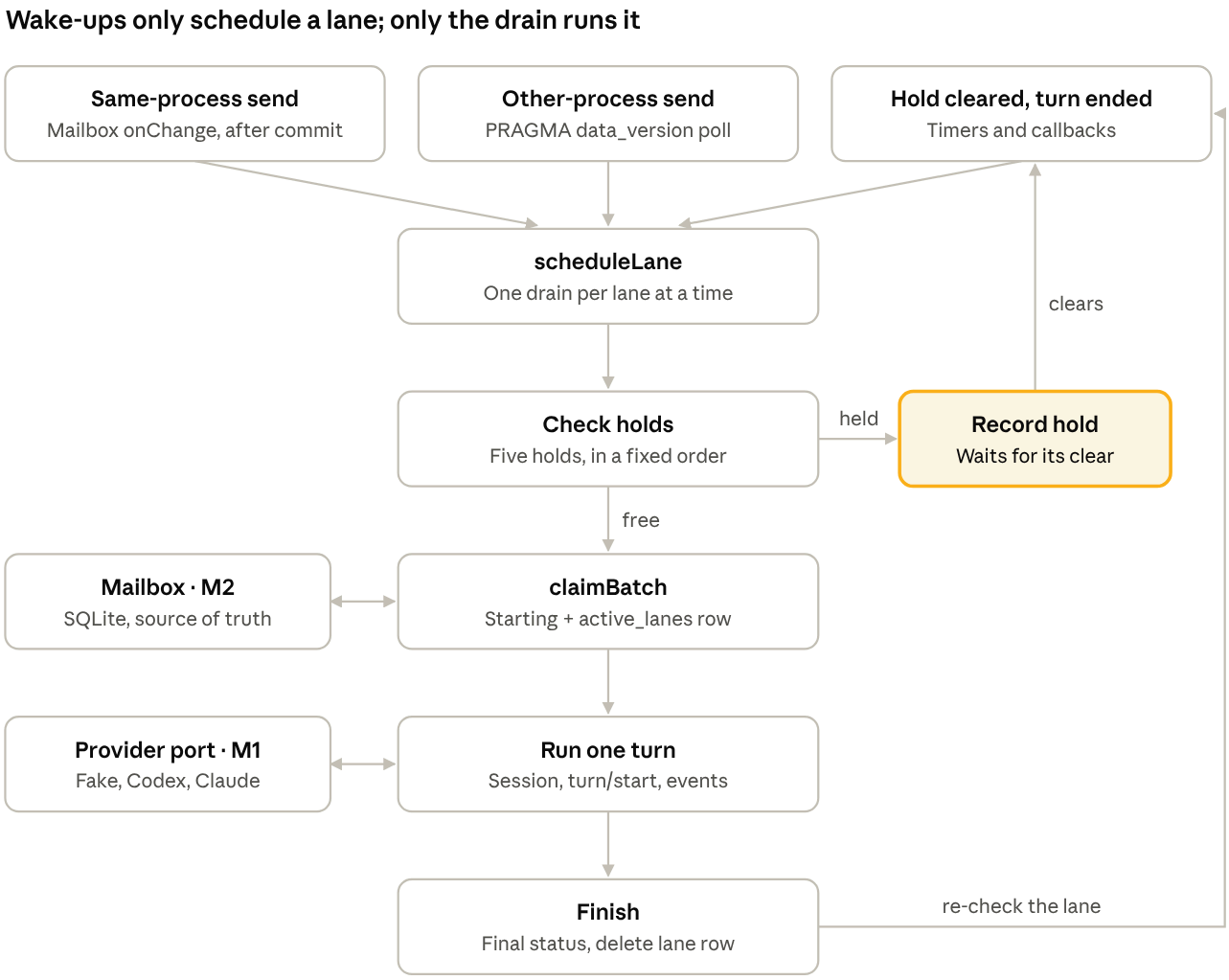
- [x] ~~Find the database the way git finds `.git`: `kru init`, the nearest ancestor `.kru/`, then `KRU_DB` and `--db` (done in M3).~~
- [x] ~~Cross-process wake-ups through `PRAGMA data_version` (done in M3).~~
- [x] ~~One active batch per lane, enforced by SQLite (done in M3).~~
- [x] ~~Re-export the plan under the name Kru.~~
- [ ] Idempotency retention before v0.1: keep keys at least 30 days, pinned while any attempt is open, with no extension on replay. The design is recorded in `m3-design.md`.
- [ ] Input size limits and secret redaction for stored provider errors before v0.1.

M3 — Lanes and the drain scheduler · Done

Status: done on 2026-11-04. Every turn runs an atomically claimed batch of deliveries. The implementation record and follow-ups are below the improvements table.

Purpose. M3 is where Kru starts running work, and the most important milestone: it decides what runs, when, and in what order. A daemon watches the mailbox, picks the next work of each lane, runs it through the M1 port and records the outcome in the M2 mailbox.

Architecture



M3 drain loop · 3 wake sources, 5 steps, 1 hold path

The key idea: wake-ups only schedule a lane, and only the drain runs it. Wake-ups can come from anywhere, any number of times; `scheduleLane` merges them into at most one drain per lane. A held lane records why and waits for that reason to clear.

Module	Owns
`src/daemon/`	`kru daemon`: the single-daemon lock, the boot sweep, wake sources and shutdown
`src/scheduler/scheduler.ts`	`scheduleLane`, the scheduled and draining sets, and the re-check after each drain
`src/scheduler/holds.ts`	Hold reasons, their order, and what clears each one

Module	Owns
`src/scheduler/runner.ts`	One turn: bind the session, start the turn, follow its events, map the outcome
`src/scheduler/slots.ts`	The global cap on concurrent turns, with person-sent work first
`src/scheduler/batch.ts`	Which queued deliveries share a turn, and the combined prompt

The scheduler depends only on the `Mailbox` port (M2), and on the `ProviderClient` port and registry (M1).

New tables (migration v2)

```
CREATE TABLE active_lanes (            -- one turn per lane, enforced by
SQLite
  agent_id   TEXT NOT NULL REFERENCES agents(id),
  lane_key   TEXT NOT NULL,            -- '' = main chat; NULL cannot be in a
primary key
  batch_id   TEXT NOT NULL,
  turn_id    TEXT,
  started_at TEXT NOT NULL,
  PRIMARY KEY (agent_id, lane_key)
);

CREATE TABLE agent_sessions (          -- which native thread continues this
lane
  agent_id         TEXT NOT NULL REFERENCES agents(id),
  lane_key         TEXT NOT NULL,
  provider         TEXT NOT NULL,
  native_thread_id TEXT NOT NULL,
  prompt_hash      TEXT NOT NULL,      -- M1 prefix hash; a change starts a
new session
  updated_at       TEXT NOT NULL,
  PRIMARY KEY (agent_id, lane_key, provider)
);

CREATE TABLE daemon_lock (             -- one daemon per database
  id             INTEGER PRIMARY KEY CHECK (id = 1),
  pid            INTEGER NOT NULL,
  start_identity TEXT NOT NULL,
```

```
  heartbeat_at   TEXT NOT NULL
);

ALTER TABLE deliveries ADD COLUMN batch_id TEXT;
-- agents.config_json gains: paused, dailyTokens
```

`claimBatch` inserts the `active_lanes` row in the same transaction as the `queued → starting` updates, so a second turn in one lane fails at the primary key even if the scheduler has a bug. `agent_sessions` keeps a lane on the same native thread across turns and daemon restarts. When the prefix hash changes, a new native session starts; until M6 adds the handoff, it starts without the earlier conversation.

Wake sources

1. Same process: the M2 `onChange` event, after commit.
2. Other processes, such as `kru send` in another terminal: the daemon reads `PRAGMA data_version` about every 100 ms. It changes only when another connection commits, so each check costs almost nothing.
3. Timers and callbacks: a turn ended, a slot freed, a budget window reset, an agent resumed.

Holds — checked in this order; each names what clears it.

Hold	Checked against	Cleared by
`paused`	`agents.config_json`	`kru agent resume`
`provider-not-ready`	The registry: CLI missing, signed out or starting	The adapter becoming ready
`lane-busy`	An `active_lanes` row for the lane	That lane's turn finishing
`budget`	Today's tokens at or above `dailyTokens` — in memory from M1 usage events, then the M4 ledger	The daily reset, or a higher budget
`slots`	Running turns at the cap (default 2)	Any turn finishing; person-headed lanes go first, then the lowest `seq`

`kru status` shows each held lane with its reason — the answer to "why is my agent doing nothing?"

Batching

`claimBatch` takes the consecutive person-sent deliveries at the head of a lane, up to 10 messages or 32 KB, in one transaction, with a shared `batch_id`. Agent and routine deliveries always run alone, because each needs its own reply context and chain. The combined prompt lists the messages in order with a fixed separator. Batching is the main token saving: each turn resends the agent's whole context, so five messages in one turn cost about one fifth of five turns.

Turn outcome to delivery state — every delivery in a batch gets the same outcome.

What happened	Delivery state
`turn/completed`: completed	`completed`
`turn/completed`: failed	`failed`
`turn/completed`: interrupted by a person	`interrupted`
Provider not ready; `turn/start` never sent	Restored to `queued` at the same place
`turn/start` refused with a definite error	`failed`
`turn/start` timed out with an unknown result	Stays `starting` until `turn/started` arrives, or a 5-minute grace ends → `interrupted`
Process exit or idle timeout during the turn	`interrupted`

Nothing is retried automatically; a person runs `kru retry`.

Daemon lifecycle

- **Start:** take `daemon_lock`. When the lock's process is gone or its start identity does not match (the M0 identity check), take it over; otherwise refuse to start.
- **Boot sweep**, a minimal M7: mark every `starting` and `running` delivery `interrupted` with "daemon restarted", and clear `active_lanes`. Without it, a crash leaves lane rows that block those lanes forever. M7 replaces it with asking the provider what happened.
- **Stop:** stop claiming, wait up to the grace period for running turns, then interrupt them. A second signal force-kills, as in M0.

Design rules

1. Wake-ups only schedule a lane; only a drain runs it.
2. One turn per lane, enforced twice: in memory by the draining set, and in SQLite by the `active_lanes` primary key.
3. Re-check after release. After the draining flag clears, ask the mailbox whether the lane has queued work and schedule it again if so.
4. Every hold names what clears it, and every held lane shows its reason in `kru status`.
5. Every turn has a delivery behind it. A turn that ends with open plan steps does not start another turn.
6. Nothing is retried automatically.
7. One daemon per database.
8. The scheduler imports the `Mailbox` and `ProviderClient` ports only, never an adapter.

Tests

#	Test	Proves
1	Five fast messages to one agent	One running turn, then one batched turn for the other four
2	Two agents with work	They run in parallel
3	A second daemon on the same database	It refuses to start
4	Each hold reason	The lane is held with that reason; clearing it resumes the lane
5	`kru send` from another process	Picked up through `data_version` within 1 second
6	A message sent in the gap before release	It runs — no lost wakeup
7	An agent or routine delivery at the head of a lane	It runs alone, not batched
8	Cap 2, three agents with work	The third waits; a person-headed lane goes first
9	`turn/start` with an unknown result	Stays `starting`; ends `interrupted` after the grace period

#	Test	Proves
10	`kill -9` of the daemon, then a restart	No `starting` or `running` deliveries, no `active_lanes` rows; the queue continues
11	A second `active_lanes` row for the same lane	Refused by the primary key
12	Count turn starts in the fake	Equal to the number of claimed batches — no turn without a delivery

Done when: all 12 tests pass in ten repeated runs.

Watch for: the lost wakeup. A message that arrives while a drain finishes has its wake-up dropped because the lane is still draining, and the drain then ends without looking again. OpenBot's `Effect.ensuring` block in `#drainLane` checks `nextQueued` again after it clears the draining flag; test 6 forces this gap. The same applies to holds: when a budget resets or a slot frees, schedule the held lanes again.

Read after: `src/backend/agent/drain-scheduler.ts` (`#scheduleLane`, `#drainLane`, `startDelivery`, `#settleUnstarted`), `src/backend/agent/turn-slots.ts`, `src/backend/agent/usage-limit-gate.ts`, `src/backend/agent/boot-recovery.ts`.

You can now (example output):

```
$ kru daemon &
$ kru send reviewer "review the latest commit"
$ kru status
reviewer  main  running  turn t-12  batch of 1
coder     main  held     budget (198,400 of 200,000 tokens today)
$ kru agent set coder --daily-tokens 400000
```

Improvements over OpenBot

Area	OpenBot	Kru M3
One turn per lane	In-memory sets only	Also a primary key in SQLite (`active_lanes`)
Wake-ups	One Electron process	Local events plus `data_version`, so the CLI and the daemon can be separate processes

Area	OpenBot	Kru M3
Automatic turns	Up to 3 plan continuations (`PlanContinuation({ limit: 3 })`)	None
Hold reasons	Computed in several places	One table; each reason names what clears it
Concurrency cap	Turn slots on hosted servers only	A global cap by default (2)
Budget	Reacts to the provider's own usage limits	Kru's own daily token budget stops work earlier

Implementation record

Code: `src/daemon/` (composition, lock, session store), `src/scheduler/` (scheduler, holds, slots, batch, session, runner), the batch operations in `src/mailbox/sqlite.ts`, and `src/store/migrations/002-scheduler.sql`. Design decisions are recorded in `m3-design.md`.

Delivered

- **CLI.** `kru init`, `kru daemon`, `kru status` (also `--json`), `agent pause`, `agent resume` and `agent set --daily-tokens`. Daemon flags: `--max-turns`, `--poll-ms`, `--idle-timeout-ms`, `--start-timeout-ms`, `--start-grace-ms` and `--shutdown-grace-ms`.
- **Atomic batches.** `claimBatch` reserves the lane and moves its deliveries to `starting` in one transaction. `active_lanes` has a unique `batch_id` and no row id. `markBatchRunning`, `finishBatch` and `restoreBatch` move every member at once; a finish must match its batch id, so a stale completion cannot release a later batch. `claimNext` still works for M2 callers but cannot bypass a reservation.
- **Batching.** Consecutive person-sent deliveries, up to 10 messages and 32,768 bytes including the separator `\n\n--- Kru message ---\n\n`. An oversized message at the head of a lane fails before any model call, so the messages behind it can run.
- **Wake-ups.** A startup scan, an after-commit change bus shared by every mailbox on one `Store`, and `PRAGMA data_version` every 100 ms, sampled before each scan on one long-lived connection.

- **Holds and slots.** The five holds in the planned order. A global cap of 2; a slot is held through session setup and the unknown-start grace. Person-headed lanes go first, then the lowest `seq`.
- **Budgets.** A start gate: a running turn can take an agent past its limit. Counted from each turn's latest `last.totalTokens`, as a difference. Kept in memory; reset at UTC midnight or a daemon restart.
- **Sessions.** Bindings keyed by agent, lane and provider, with the M1 `prompt_hash` and a `config_hash` (prefix, provider, model, cwd, permissions). An unchanged binding is reused; after a restart the runner tries `thread/resume`. A setup failure restores the batch and holds the provider without sending `turn/start`.
- **Outcomes.** As in the table above. Listeners are installed before `turn/start`; a late `turn/started` confirms the original batch. Output, progress and tool activity reset the idle timer; a running native command pauses it. When an outcome stays unknown, the runner force-stops the provider runtime before releasing the lane.
- **Daemon.** The lock uses PID and the M0 start identity, with a heartbeat every second and a status snapshot for `kru status`. A live matching PID always keeps the lock. The boot sweep marks `starting` and `running` deliveries `interrupted`, clears reservations and discards their session bindings. Shutdown waits up to 5 seconds, then interrupts; a second signal force-stops.

Changes from the plan

Plan	Built	Why
`daemon_lock` with a heartbeat	Also a status snapshot	Lets another CLI process show hold reasons and token totals
`agent_sessions` with `prompt_hash`	Also `config_hash`	A model, cwd or permission change also starts a new session
An unknown outcome ends `interrupted`	Also force-stops the provider runtime	Stops an unknown turn from running on unseen
Budget counted from usage events	Counted from the latest snapshot as a difference	Repeated usage events are not added twice

Verification

- The 12 numbered tests in `test/scheduler.test.ts`, with real CLI sends from another process, competing batch claims, a refused second daemon, a forced lost-wakeup gap and a real `SIGKILL` and restart.
- 43 scheduler tests run ten times: 430 passes, no failures. Frozen v1 and v2 fixtures check migration parity and rollback.
- Full check: TypeScript passes; 203 tests pass, 9 opt-in tests skipped. No model or API key is needed.

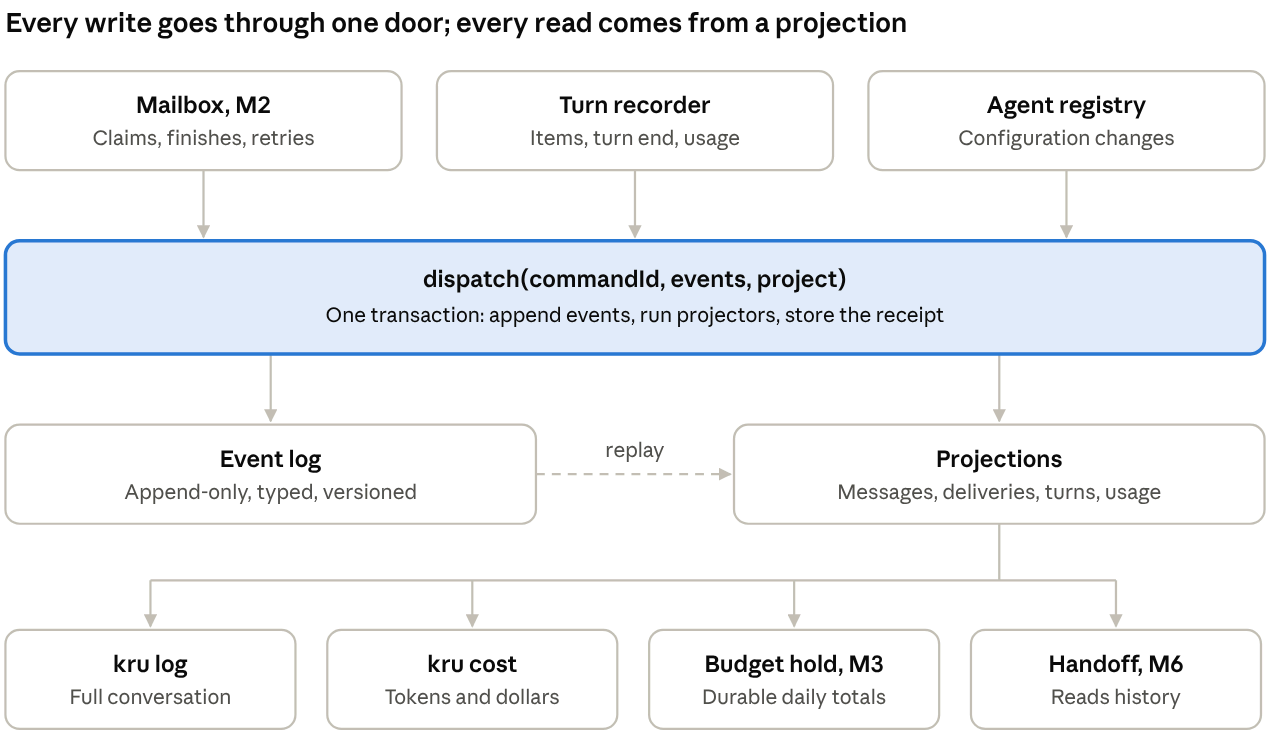
Follow-ups

- [x] ~~Targeted recovery for an unknown Codex turn: abandoned threads get tombstones, and a liveness probe restarts the shared process only when it stops answering, so healthy agents are not interrupted. Claude keeps per-thread process termination.~~
- [x] ~~Heartbeat every 10 seconds; the status snapshot is written only when it changes.~~
- [x] ~~Durable budget counters — moved to M4, where the cost ledger provides them.~~
- [x] ~~Budget days follow local time, with a `--budget-timezone` override.~~
- [x] ~~The boot sweep keeps session bindings and tries resume before replacing a missing cache.~~
- [x] ~~Mailbox adapters and agent registries on one `Store` publish on one change bus, so configuration changes wake the scheduler.~~
- [x] ~~Tag v0.0.5.~~

M4 — Event log and projections

Purpose. M4 gives Kru a memory. Everything that happens becomes an event in an append-only log, and the tables Kru reads become projections of those events. M4 also adds the cost ledger: every turn records its tokens and its dollars. The M1 benchmark showed why dollars matter: the lean profile used 34% fewer tokens but cost about 51% more at list prices, because it produced more output and more cache writes.

Architecture



M4 write and read paths · 3 writers, one dispatch, 4 readers

Every write goes through one door, `dispatch`, and every read comes from a projection. When a projection is wrong or its schema changes, replaying the log rebuilds it.

Domain state and operational state

Event sourcing makes every write two writes and the log only grows, so M4 splits Kru's state in two:

Kind	Examples	Event-sourced
Domain state: what happened and what it cost	Messages, delivery transitions, turns, items, usage, agent configuration	Yes; replay rebuilds it
Operational state: who holds what now	`active_lanes`, `daemon_lock`, `agent_sessions`	No; plain tables that the boot sweep resets

Module	Owns
`src/store/events.ts`	`dispatch`, the event table, command receipts, typed and versioned payloads

Module	Owns
`src/store/projectors/`	One projector per projection (mailbox, conversation, usage), each a function from an event to table writes
`src/turns/recorder.ts`	Provider events from the M3 runner; live deltas from memory; items and turn ends written as events
`src/cost/prices.ts`	The price table and the cost of each turn
`src/cli/`	`kru log`, `kru cost` and `kru replay --verify`

Events

Event	Written by	Updates
`message.enqueued`	Mailbox	`messages`, `deliveries`
`delivery.claimed`, `running`, `restored`, `finished`, `cancelled`, `retried`	Mailbox	`deliveries`
`turn.started`	Recorder	`turns`: agent, lane, batch, provider, model, profile, prefix hash
`item.completed`	Recorder	`thread_messages`: assistant text, tool calls, commands, file changes
`turn.completed`	Recorder	`turns`: status, error, duration
`usage.recorded`	Recorder	`usage_records`: tokens, cost, price id, cause, chain
`agent.configured`	Registry	`agents`

Message deltas are never events; there are too many. They stream from memory to live listeners such as `kru log --follow`, and only the final text of each item is stored. Optionally, the recorder saves a partial snapshot about every 2 seconds to a scratch table, so another process can show live text and a crash leaves the partial text for M7.

A turn ends in one transaction

When a turn ends, one `dispatch` writes `delivery.finished` for each batch member, the remaining `item.completed` events, `turn.completed` and `usage.recorded`, and removes the lane reservation. All of it commits or none of it does: no delivery is ever `completed` without its answer or its cost. `dispatch` becomes the only transaction owner, and the M3 batch operations become commands that run inside it, as the M2 notes anticipated.

The cost ledger

```
CREATE TABLE prices (
  id TEXT PRIMARY KEY, provider TEXT NOT NULL, model TEXT NOT NULL,
  input_usd REAL NOT NULL, cache_write_5m_usd REAL, cache_write_1h_usd REAL,
  cache_read_usd REAL, output_usd REAL NOT NULL,      -- per million tokens
  effective_from TEXT NOT NULL, source_url TEXT NOT NULL
);

CREATE TABLE usage_records (
  turn_id TEXT PRIMARY KEY, agent_id TEXT NOT NULL, lane_key TEXT NOT NULL,
  cause TEXT NOT NULL, chain_id TEXT, provider TEXT NOT NULL, model TEXT NOT
NULL,
  input_total INTEGER NOT NULL, cached_read INTEGER NOT NULL,
  cache_write INTEGER NOT NULL, cache_write_ttl TEXT, output INTEGER NOT
NULL,
  estimated INTEGER NOT NULL, price_id TEXT REFERENCES prices(id),
  cost_usd REAL,                      -- NULL when no price is known, never 0
  recorded_at TEXT NOT NULL           -- UTC
);
CREATE INDEX usage_by_agent_time ON usage_records (agent_id, recorded_at);
```

- Cost is computed when the turn is recorded and stored with its price id, so a later price change never rewrites history.
- An unknown price stores `NULL`. `kru cost` shows unpriced turns on their own line and never counts them as free.
- The cache-write lifetime is stored: a 1-hour cache write costs about 60% more than a 5-minute one.
- Budgets come in tokens and dollars, and the M3 budget hold checks both. Daily totals are a range query on `recorded_at` from the start of the local day, so a change of budget timezone never moves past usage to another day. Totals load from the table at startup, which ends the reset on restart.

Migration v3

The database already holds M2 and M3 messages and deliveries. Migration v3 writes genesis events (`message.imported`, `delivery.imported`) from the current rows, in `seq` order, so a replay reproduces today's state. Its test migrates the frozen v2 fixture, replays every event into an empty database and compares every projection row by row.

Design rules

1. `dispatch` is the only transaction owner. Projectors never open a transaction; OpenBot's `src/backend/AGENTS.md` states the same rule and what breaks without it.
2. Projectors are deterministic: no clock, no random ids, no network. Every value they write comes from the event payload.
3. Domain state is event-sourced; operational state is not.
4. A turn ends in one `dispatch`: final statuses, items, turn end, usage and lane release.
5. Message deltas are never events.
6. Shipped event shapes are frozen, like migrations. A change is a new payload version plus an upcaster that converts old events when they are read.
7. Command receipts make commands idempotent: a repeat returns the stored result and appends nothing.
8. Cost is stored with its price id; an unknown price is `NULL`, never 0.
9. Budgets are in tokens and dollars, counted by UTC timestamp ranges.

Tests

#	Test	Proves
1	`kru replay --verify` on a busy database	Every projection rebuilds identically
2	Replay into two empty databases	Both are identical, timestamps included
3	A command run twice	The stored result returns; no events are appended
4	A failure injected halfway through a turn end	Nothing commits: no status, items, usage or lane release
5	Daemon restart	`kru log` shows the full conversation
6	`kru cost` totals	Equal the sum of the usage events
7	A model with no known price	`cost_usd` is `NULL`; the turn shows as unpriced

#	Test	Proves
8	A price change after a turn	That turn's cost does not change
9	Daemon restart with a spent budget	The budget stays spent, in tokens and in dollars
10	A budget timezone change	Past usage stays on its day
11	An event stored in an old payload version	Its upcaster reads it
12	The frozen v2 fixture	Migrates, replays and matches row by row

Done when: all 12 tests pass in ten repeated runs.

Watch for: usage can arrive several times for one turn; store the turn's final value, never the sum. Import genesis events in `seq` order, or replay rebuilds a different queue.

Read after: `src/backend/database/database-core.ts` (`dispatch`), `src/backend/database/thread-replay.ts`, `src/backend/agent/delta-buffer.ts`, `handleNotification` in `src/backend/agent/turn-lifecycle.ts`, `src/backend/agent/usage-collection.ts`, `src/backend/agent/usage-pricing.ts`, `src/backend/database/agent-usage.ts`, and the transaction rules under Size in `src/backend/AGENTS.md`.

You can now (example output):

```
$ kru log reviewer --follow
$ kru cost --by agent --since 7d
agent      turns   input     cached    output   cost
reviewer      42   510,200   431,900   38,120   $1.12
coder         17   205,400   160,010   11,300   $0.61
unpriced       2                                 n/a
$ kru replay --verify
replayed 1,284 events; 6 projections match
```

Improvements over OpenBot

OpenBot uses the same core design — `orchestration_events`, projection tables and command receipts in one transaction — so this part learns from it directly. The differences:

Area	OpenBot	Kru M4
Replay check	Rebuilds one thread's projection (`ThreadReplay`)	`kru replay --verify` rebuilds every projection and compares
What the log holds	Not stated as a rule	An explicit split between domain and operational state
Cost	Token counts, with a price table used for estimates (`usage-pricing.ts`)	A stored cost per turn with its price id; unpriced turns shown, never $0; dollar budgets
Event changes	`schema_migrations` covers tables	Payload versions and upcasters cover events too
Existing data	Built with the event log from the start	Genesis events import earlier rows, checked by a replay test

M5 — Tools and agent-to-agent messages

- **Build:** route the provider's `item/tool/call` requests. Implement `send_message` as `mailbox.enqueue({ sender: { kind: "agent" } })` with the key `threadId + turnId + callId` . Add `list_agents` . These two are the only Kru tools in M5; no tool discovery.
- **Build — token controls:**
  - **Hop limit.** Every delivery carries a chain id and a hop count. A message an agent sends joins the chain of the turn that sent it, one hop higher. Past `maxHops` (default 4), `send_message` returns a clear tool error, so the model stops instead of passing work around.
  - **Replies are opt-in.** `expectsReply` defaults to false. An answer to a request never asks for an answer back.
  - **Visible cost.** Each chain's turns carry the chain id into the M4 ledger, so `kru cost --by chain` shows what one exchange cost.
- **Read after:** `handle` and `#sendMessage` in `src/backend/agent/openbot-tool-router.ts` , `src/backend/agent/tool-call-idempotency.ts` , `src/backend/agent/delivery-content.ts` .
- **Done when:**

  - agent A asks agent B, A's turn ends without waiting, and B's answer reaches A as a new delivery;
  - an acknowledgement loop caused on purpose (A thanks B, B thanks A) stops, because replies are off by default;
  - a chain that reaches `maxHops` ends with a tool error and starts no further turns;
  - `kru cost --by chain` shows the turns and tokens of one A↔B exchange.
- **Token note:** in OpenBot every message and every reply is a full turn for the recipient, with its whole context. One question between two agents costs at least two turns. The hop limit and opt-in replies bound how many turns one request can cause.
- **You can now:** create a reviewer agent and a coder agent that hand work to each other, within a known cost. Release **v0.1**.

M6 — Threads apart from provider sessions

- **Build:** your `threads` table and a `provider_sessions` binding table. On a provider switch or a missing session, start a new native session and prepend a handoff packet under a character budget. Add a `history_search` tool for what the handoff cut.
- **Read after:** `ensureThread` in `src/backend/agent/thread-lifecycle.ts`, `src/backend/agent/handoff-history.ts`, "History retrieval" in `docs/architecture/providers.md`.
- **Done when:** after ten turns on Claude you switch the agent to Codex, and it still answers questions about the first turn.
- **You can now:** `kru agent set <agent> --provider codex` with no lost context.

M7 — Crash recovery

- **Build:** at boot, before providers start, mark every `starting` or `running` delivery and every streaming message `interrupted`. Never retry them automatically. Let queued work continue.
- **Read after:** `src/backend/agent/boot-recovery.ts`.
- **Done when:** you `kill -9` the daemon during a turn of the stalling fake, start it again, and the state is consistent and the queue continues.
- **You can now:** run Kru as a service and trust it. Release **v0.2**.

Road to v1.0

After M7 the engine is complete. v1.0 adds what makes Kru useful to people who never read its code: an API, scheduled work, and limits that keep it safe on a shared machine.

Feature	What it adds	Read in OpenBot
HTTP + SSE API	Send, list, stream and cancel over HTTP, so any tool can drive Kru	`src/main/team-api-server.ts` (shape only)
Routines	A timer or webhook that enqueues a delivery; no new execution path	`src/backend/agent/routine-scheduler.ts`, `docs/architecture/events.md`
Context compaction	Ask the provider to compact near 80% of the window; rotate the session when it cannot	`src/backend/agent/context-compaction.ts`
Global turn slots	A cap on concurrent turns, people's messages first	`src/backend/agent/turn-slots.ts`
Deferred tool discovery	`tool_search` and `tool_call` keep the always-loaded tool list small	`src/backend/agent/tool-catalog.ts`
Agent memory	`remember`, `search_memories`, `forget_memory` tools on a SQLite table	`src/backend/agent/agent-memories.ts`
Workspace sandbox	Each agent writes only to its own folder plus allowed roots	`src/backend/agent/workspace-sandbox.ts`
Usage limits	Hold a lane when the provider reports a limit; resume when it resets	`src/backend/agent/usage-limit-gate.ts`

Build the API first. It turns Kru from a CLI into a platform, and every later feature can be tested through it.

Release checklist

People adopt a tool they can install and see working in five minutes. Do these before you announce v0.1.

- [ ] Check that the name "kru" is free on npm and GitHub; short names are often taken, so a scoped package such as @<your-name>/kru is the fallback
- [ ] Add an MIT `LICENSE` and a note that the project is inspired by OpenBot and shares no code with it
- [ ] One-line install: `bunx kru` or `npm i -g kru`
- [ ] README with a 30-second demo GIF: two agents hand work to each other
- [ ] A quick start that works with the fake provider, so a reader needs no API key to try it
- [ ] `examples/`: coder + reviewer pair, a Slack bot over the HTTP API, a nightly routine
- [ ] `docs/architecture.md` with the lifecycle diagram from this doc
- [ ] CI that runs every test against the fake providers
- [ ] A security note: agents run with the provider's own permissions; Kru adds no sandbox until v1.0
- [ ] Secrets are redacted from logs and stored error messages
- [ ] A `CHANGELOG.md` and semantic version tags
- [ ] A write-up of what you learned at each milestone, for your portfolio

How to read the OpenBot source

Read the docs for *why*, the tests for *what*, and the history for *how they got there*. File paths in this doc are from nightly-labs/openbot at commit `0b311bf` (v0.57.0).

1. Clone the full history, not a shallow copy: `git clone https://github.com/nightly-labs/openbot.git`.
2. Read `docs/architecture/structure.md`, `providers.md` and `src/backend/AGENTS.md` before any code. They state the rules the code obeys.
3. Read `docs/glossary.md`. It maps product words (thread, side thread, delivery, lane) to code names.
4. Treat `scripts/*-e2e.ts` and `scripts/provider-fakes/agents/*.ts` as behaviour specs. `stall.ts` and `session-errors.ts` show the failures they guard against.
5. Use `git log -p -- <file>` on the file of your current milestone. Bug fixes show the edge cases you will hit.

6. Follow one message by searching method names: `enqueue` → `scheduleDrain` → `startDelivery` → `"turn/start"` → `handleNotification` → `completeTurn`.
7. Skip `src/renderer`, `src/main` and `apps/` at first. They are product surface, not engine.

Sources

- nightly-labs/openbot on GitHub — repository read at commit `0b311bf`, 2026-11-01